

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Complementary events

1. If the probability of a player winning a game is 0.79, then the probability of his losing the same game is: (a) 1.79 (b) 0.31 (c) 0.21% (d) 0.21 **CBSE 2024, 1 mark**

SOLUTION

$$P(\text{losing}) = 1 - P(\text{winning}) = 1 - 0.79 = 0.21$$

Answer: (d)

2 Theoretical probability

2. From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting at random a prime number from the remaining is: (a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{7}$ (d) $\frac{2}{7}$

CBSE 2024, 1 mark

SOLUTION

Removing the even numbers 4 and 16 leaves 1, 7, 9, 21, 25: five numbers.

The only prime among them is 7 (1 is not prime; $9 = 3^2$, $21 = 3 \times 7$, $25 = 5^2$).

$$\text{Probability} = \frac{1}{5}$$

Answer: (b)

3 Two dice

3. Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5, is: (a) $\frac{7}{36}$ (b) $\frac{11}{36}$ (c) $\frac{5}{36}$ (d) $\frac{4}{9}$ **CBSE 2024, 1 mark**

SOLUTION

Sum 2: (1, 1). Sum 3: (1, 2), (2, 1). Sum 5: (1, 4), (2, 3), (3, 2), (4, 1).

Favourable outcomes = $1 + 2 + 4 = 7$ out of 36.

Answer: (a)

4 Playing cards

4. In a pack of 52 playing cards one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is queen of heart, if the lost card is a black card.

CBSE 2024, 2 marks

SOLUTION

One black card is lost, so 51 cards remain.

The queen of hearts is a red card, so it is still in the pack.

Answer: $\frac{1}{51}$